

Memory Cleaner

User Manual · Version 1.0

A free, portable tool that frees up RAM on Windows using the operating system's own memory-management functions.

For Windows 10 and Windows 11 · No installation required

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> **How Windows uses memory.** Windows deliberately keeps recently used files and program data in RAM (the *standby list*) so they open faster next time. That is why Task Manager can show high memory use even when little is running. Memory Cleaner lets you hand that memory back on demand, which helps most when a game or heavy program needs a large block of free RAM right away.

1 Getting started

Memory Cleaner is portable: it is a single program file that runs from any folder and never installs anything.

What you need

- Windows 10 or Windows 11 (Windows 7 and 8.1 also work, with fewer memory areas available).
- .NET Framework 4.8, which is already built into Windows 10 (May 2019 update and later) and Windows 11.
- An administrator account, or the password for one. Clearing system memory lists requires administrator rights.

What's in the download

File	Purpose
`MemoryCleaner.exe`	The program itself (about 76 KB).
`README.txt`	A one-page quick reference.
`MemoryCleaner.ini`	Created automatically the first time you run the program. Holds your settings.

First run

1. **Unzip** `MemoryCleaner-portable.zip` into a folder you'll keep, for example `C:\Tools\MemoryCleaner`. Running it straight from inside the zip file means your settings can't be saved.
2. **Double-click** `MemoryCleaner.exe`.
3. **If a blue "Windows protected your PC" screen appears**, click **More info**, then **Run anyway**. This appears because the program isn't digitally signed; it only happens the first time.
4. **Approve the User Account Control prompt** by clicking **Yes**. Memory Cleaner asks for this each time you start it by hand.
5. The main window opens. Click **Optimize** to try it.

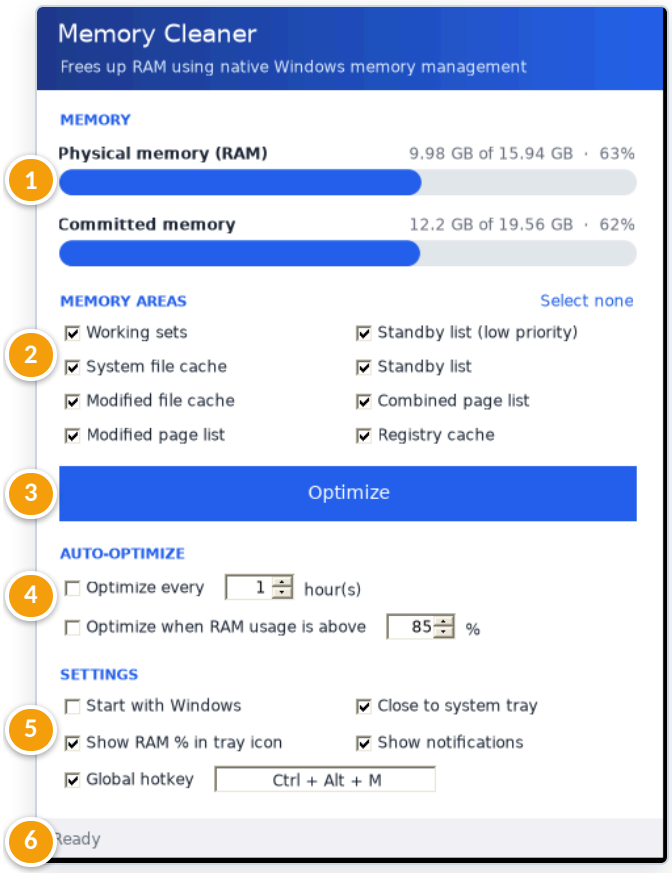
> **Don't move the file to a protected location without admin rights.** Memory Cleaner saves its settings next to the program. If that folder isn't writable, it falls back to `%AppData%\MemoryCleaner\` automatically, so nothing breaks, but the settings won't travel with the program on a USB stick.

Only one copy runs at a time

If Memory Cleaner is already running (for example, hidden in the tray) and you start it again, the new copy simply brings the existing window to the front and closes itself.

2 A tour of the main window

Everything lives on one screen. Changes are saved the moment you make them; there is no Save or Apply button.



1. **Memory bars.** *Physical memory* is your installed RAM. *Committed memory* is what programs have reserved across RAM and the page file, measured against the most Windows can hand out. Both refresh every second. The bars turn amber at 75% and red at 90%.
2. **Memory areas.** Tick the areas to clear when you optimize. Hover over any area for a short explanation. **Select all** / **Select none** toggles every box. Areas your version of Windows doesn't support are greyed out. See chapter 3.
3. **Optimize.** Clears the ticked areas now. The button reads *Optimizing…* while it works, usually for a second or two.
4. **Auto-optimize.** Clean on a timer and/or when RAM gets too full. See chapter 5.
5. **Settings.** Startup, tray behavior, notifications and the global hotkey. See chapter 7.
6. **Status bar.** Shows the result of the last clean, for example *Freed 2.4 GB at 9:14 PM*, and on the right the total freed since the program started.

3 Memory areas explained

Each area is a different part of Windows' memory system. They are cleared in the order below, which gives the best result.

Area	What it does	Frees
Working sets	Asks every running program to give back memory it isn't actively using. The data isn't lost; it moves to the standby or modified list and is brought back instantly if the program needs it.	Moderate
System file cache	Shrinks the memory Windows uses to cache file-system data.	Small to moderate
Modified file cache	Writes any data waiting in the write cache of every internal drive to disk. Mostly a housekeeping step for the next two areas.	Small
Modified page list	Writes changed memory pages out to disk so they become reusable.	Small to moderate
Standby list (low priority)	Empties only the least important cached data. A gentle option that keeps your most-used files cached.	Moderate
Standby list	Empties the *entire* cache of recently used files and programs, turning it all into free memory. When this is ticked, the low-priority step is skipped because it's already covered.	**Large**
Combined page list Windows 10 and later	Finds memory pages with identical contents and keeps just one copy.	Small
Registry cache Windows 8.1 and later	Writes registry data held in memory back to disk and releases it.	Small

Which areas should I tick?

If you want...	Tick
The most free RAM right now (for example, before starting a game)	Everything (the default).
Some relief without slowing anything down	Everything **except** *Standby list*, and leave *Standby list (low priority)* ticked.
Frequent automatic cleaning	The gentle set above. Emptying the full standby list every few minutes works against Windows' caching.

> **A note on the full standby list.** Clearing it produces the biggest number, but that memory was a cache of files you recently used. Windows re-reads them from disk the next time they are needed, so apps you had open may feel briefly slower right after a clean. This is normal and harmless.

4 Optimizing memory

There are four ways to start a clean. They all use the areas ticked in the main window.

Method	How
Optimize button	Click **Optimize** in the main window.
Global hotkey	Press Ctrl + Alt + M from any program, even a full-screen game. You can change the keys; see chapter 7.
Tray menu	Right-click the tray icon and choose **Optimize now**.
Automatically	On a timer or when RAM is too full; see chapter 5.

Reading the results

- **Status bar:** shows how much memory was freed and when, for example *Freed 1.8 GB at 10:02 PM (hotkey)*. Cleans you didn't start with the button show what triggered them in brackets.
- **Session total:** the right side of the status bar adds up everything freed since Memory Cleaner started.
- **Notifications:** when the window is hidden, or the clean was automatic or from the hotkey, a Windows notification pops up with the amount freed. You can turn these off.

> **How "freed" is measured.** Memory Cleaner reads the amount of available RAM just before and just after the clean and reports the difference. If your RAM was already mostly free, or other programs grabbed memory during the clean, the figure can be small or zero. That doesn't mean it failed.

Warnings

If an area couldn't be cleared, the status bar turns amber and adds *1 warning (hover)*. Hold your mouse over the status bar to see which area failed and why. The other areas are still cleared normally. See chapter 10 for common causes.

5 Automatic optimization

Memory Cleaner can clean for you in the background while it runs, including when it's hidden in the tray.

Optimize every N hours

Tick **Optimize every** and choose 1 to 24 hours. The timer counts from the *last* clean of any kind, so if you clean by hand, the next scheduled clean is pushed back accordingly. Changing the number restarts the countdown.

Optimize when RAM usage is above N%

Tick **Optimize when RAM usage is above** and choose a threshold between 50% and 99% (default 85%). Memory Cleaner checks once per second. When physical memory use reaches the threshold, it cleans.

To avoid cleaning over and over, this trigger waits at least **5 minutes** after any clean before it can fire again, even if usage stays high.

Using both together

Both options can be on at once. A good combination for everyday use:

- Every **2 hours**, plus when usage goes above **90%**.
- Memory areas: the gentle set from chapter 3 (everything except the full *Standby list*).
- **Start with Windows** ticked so it's always running.

> **Automatic cleaning only happens while the program is running.** Closing the window keeps it running in the tray (unless you turned that off). Choosing **Exit** from the tray menu stops it completely.

6 The system tray icon

Memory Cleaner places an icon in the notification area at the bottom-right of the taskbar.

Live RAM gauge

With **Show RAM % in tray icon** ticked, the icon shows your current RAM usage as a number and changes color as memory fills up:



Blue	Amber	Red
below 75%	75–89%	90% and up

The icon shows at most 99. Untick the setting to show the normal Memory Cleaner icon instead. Hovering over the icon shows a tooltip with the exact amount in use.

> **Can't see the icon?** Windows may hide new tray icons behind the `^` arrow. On Windows 11, open *Settings → Personalization → Taskbar → Other system tray icons* and switch on Memory Cleaner to keep it always visible.

Mouse actions

Action	Result
Left-click	Opens the main window.
Right-click	Opens the menu: **Open Memory Cleaner**, **Optimize now**, and **Exit**.

Closing vs. exiting

With **Close to system tray** ticked (the default), clicking the window's `×` only hides the window; the program keeps monitoring, auto-optimizing and listening for the hotkey. The first time you do this, a notification reminds you it's still running. To quit completely, right-click the tray icon and choose **Exit**.

7 Settings

All settings are in the bottom section of the main window and take effect immediately.

Start with Windows

Starts Memory Cleaner hidden in the tray about 10 seconds after you sign in, **without** a User Account Control prompt.

Windows doesn't allow programs that need administrator rights to start through the usual startup list, so Memory Cleaner creates a task named `MemoryCleaner` in **Task Scheduler** instead. The task runs with the highest privileges your account has, never times out, and also runs on battery power. Unticking the box deletes the task.

> **Moved the program to a new folder?** Just run it once from the new location. If startup is enabled, Memory Cleaner updates the task to point at its new location automatically.

Close to system tray

When ticked, closing the window hides it instead of quitting (see chapter 6). Untick it if you want the ✕ button to exit the program.

Show RAM % in tray icon

Switches the tray icon between the live percentage gauge and the standard program icon.

Show notifications

Shows a Windows notification after automatic, hotkey and tray cleans, and whenever you clean while the window is hidden. A clean started from the **Optimize** button with the window open doesn't notify you, because the status bar already shows the result.

Global hotkey

Tick **Global hotkey** to clean from anywhere with a key combination. The default is Ctrl + Alt + M.

1. Click inside the hotkey box.
2. Press your new combination. It must include Ctrl or Alt (you can add Shift), for example Ctrl + Shift + F12.
3. Click anywhere else. The new hotkey is active right away.

If the text in the box turns **red**, another program already uses that combination. Hover over the box to confirm, then pick a different one.

8 Command-line options

For power users who want to clean from scripts, desktop shortcuts or Task Scheduler.

Option	What it does
`/optimize`	Cleans once using the memory areas saved in your settings, then exits. No window or tray icon appears. It returns exit code `0` if every area succeeded and `1` if any reported a problem.
`/tray`	Starts the program hidden in the tray. This is what **Start with Windows** uses.

Options can also be written with a dash (`-optimize`) and are not case-sensitive.

Example: a one-click "clean RAM" desktop shortcut

1. Right-click `MemoryCleaner.exe` and choose **Create shortcut** (on Windows 11 it's under **Show more options**).
2. Right-click the new shortcut and choose **Properties**.
3. In **Target**, add `/optimize` after the closing quote, for example:

```
"C:\Tools\MemoryCleaner\MemoryCleaner.exe" /optimize
```

Double-clicking the shortcut now cleans your RAM in the background (after the admin prompt) without opening the window.

Example: from a batch file

```
@echo off
"C:\Tools\MemoryCleaner\MemoryCleaner.exe" /optimize
if errorlevel 1 echo Some memory areas could not be cleared.
```

Because the program needs administrator rights, run the batch file from an elevated Command Prompt or from a Task Scheduler task set to *Run with highest privileges* to avoid the prompt.

9 The settings file

You never need to edit this file by hand, but it's plain text if you want to copy settings between computers.

Settings are stored in `MemoryCleaner.ini` in the same folder as the program. If that folder can't be written to, the file is kept in `%AppData%\MemoryCleaner\` instead. Close Memory Cleaner before editing it, or your changes may be overwritten.

Key	Default	Meaning
`Areas`	`255`	Which memory areas are ticked, as the sum of the values in the table below.
`AutoInterval`	`0`	`1` = optimize every N hours is on.
`IntervalHours`	`1`	Hours between timed cleans (1–24).
`AutoUsage`	`0`	`1` = optimize above the usage threshold is on.
`UsageThreshold`	`85`	RAM usage percentage that triggers a clean (50–99).
`CloseToTray`	`1`	`1` = closing the window hides it to the tray.
`TrayShowUsage`	`1`	`1` = show RAM % in the tray icon.
`Notifications`	`1`	`1` = show notifications.
`HotkeyEnabled`	`1`	`1` = the global hotkey is on.
`Hotkey`	`393293`	The key combination in .NET `Keys` format. Easier to set from the app.

Start with Windows isn't stored here; it's read from Task Scheduler each time the program starts.

Area values

Area	Value	Area	Value
Working sets	1	Standby list (low priority)	16
System file cache	2	Standby list	32
Modified file cache	4	Combined page list	64
Modified page list	8	Registry cache	128

For example, the "gentle" set from chapter 3 (everything except the full standby list) is 255 − 32 = `Areas=223`.

10 Troubleshooting & FAQ

Answers to the most common questions.

Windows says "Windows protected your PC" and won't start it.

Click **More info**, then **Run anyway**. The warning appears because the program isn't signed with a paid certificate. If your antivirus quarantines it, add an exception for the Memory Cleaner folder; tools that change system memory are sometimes flagged by overly cautious scanners.

The status bar says "Not running as administrator".

Memory Cleaner was started without administrator rights, so most areas can't be cleared. Close it and start it again, choosing **Yes** at the prompt. If your account isn't an administrator, you'll need an administrator's password.

An area shows a warning like "privilege not held".

Windows didn't grant the program the rights it needs, usually because of a company or school policy. The other areas still work. Untick the failing area to stop the warning.

"Modified file cache: could not flush D:"

A drive was locked, busy, or doesn't support flushing (for example, a locked BitLocker drive or certain virtual drives). This is harmless. Untick *Modified file cache* if it happens every time.

Some areas are greyed out.

Your version of Windows doesn't support them. *Combined page list* needs Windows 10 or later; *Registry cache* needs Windows 8.1 or later.

It freed 0 MB or very little.

There wasn't much to release, or programs took memory back while it was cleaning. Try ticking *Standby list*, which frees the most. See the note on measuring in chapter 4.

My PC feels slower right after a clean.

Clearing the full standby list removes Windows' file cache, so recently used files are read from disk again. It settles within a minute. For regular or automatic cleaning, untick *Standby list* and use *Standby list (low priority)*.

The hotkey doesn't work.

Check that **Global hotkey** is ticked and the text in the box isn't red. If it is red, another program owns that combination; choose a different one. The hotkey only works while Memory Cleaner is running (window open or in the tray).

It didn't start with Windows.

Look in the tray first, as it starts hidden about 10 seconds after sign-in and may be behind the `^` arrow. If it isn't there, open *Task Scheduler*, find the `MemoryCleaner` task in the Task Scheduler Library, and check that it's enabled and points to the right file. Unticking and re-ticking **Start with Windows** recreates it.

Is this safe? Will I lose data?

Yes, it's safe. Memory Cleaner only calls functions built into Windows for managing memory. It never closes programs or deletes files. Anything written to disk during a clean (modified pages, drive caches) would have been written by Windows anyway.

Does it need to stay open?

Only if you use automatic cleaning or the hotkey. For an occasional manual clean, open it, click **Optimize**, and exit.

11 Removing Memory Cleaner

Because nothing is installed, removing it takes two steps.

1. **If you enabled Start with Windows, untick it first.** This deletes the `MemoryCleaner` task from Task Scheduler. (You can also delete the task yourself in Task Scheduler.)
2. **Exit the program** from the tray menu, then **delete the folder** containing `MemoryCleaner.exe` and `MemoryCleaner.ini`.

If the settings file ended up in the fallback location, also delete `%AppData%\MemoryCleaner\`. Paste that path into File Explorer's address bar to go straight there.

> **Source code.** The full C# source code is included in `MemoryCleaner-source.zip`. On a Windows PC with the .NET SDK installed, run `dotnet build -c Release` in that folder to build your own copy.